

Mohammed Alshowaikhat

✉ mshowaikhat@gmail.com ☎ +966556777921 📍 Saihat, Saudi Arabia 🔗 [linkedin.com/in/m-showaikhat](https://www.linkedin.com/in/m-showaikhat)
🔗 Portfolio: mshowaikhat.github.io

SUMMARY

Senior Computer Science at KFUPM specializing in cloud-native backends and applied GenAI. Built a bilingual Arabic-English appointment agent on AWS Bedrock spanning web, WhatsApp, and voice channels as an internal POC, with serverless infrastructure managed via CDK. AWS and GCP certified, comfortable owning systems from architecture to delivery.

EDUCATION

King Fahd University of Petroleum & Minerals (KFUPM), 2021 – 2026 | Dhahran, Saudi Arabia
Bachelor's degree in Computer Science
Third Honor in Computer Science, GPA: 3.195/4.0
Relevant Courses: Cloud Applications Engineering, Web Engineering & Development, Network Management & Security

PROFESSIONAL EXPERIENCE

Octedia, Software Development Intern 08/2025 – 12/2025 | Dammam, Saudi Arabia

- Sole engineer on a multi-channel GenAI appointment booking system (web, WhatsApp, voice) with bilingual Arabic-English support for breast cancer screening; delivered as an internal POC.
- Architected a serverless backend on AWS Lambda, DynamoDB, API Gateway, and SNS, orchestrated by Bedrock Agents with custom action groups; provisioned via AWS CDK for reproducible infrastructure-as-code deployments.
- Built a bilingual web chatbot using Next.js, FastAPI, and AWS Bedrock, with real-time voice via AWS Transcribe and Polly over WebSocket; achieved 60–80% latency reduction.
- Extended the system to phone calls by building a custom Lambda adapter that exposes AWS Bedrock Agents through an OpenAI-compatible API, allowing Telnex telephony to drive the existing agent backend with no infrastructure changes.

PROJECTS

Air Traffic Control Simulator with AI Voice Integration, 2026
Multidisciplinary Senior Capstone, KFUPM

- Led the AI module of a 6-person capstone team building an ATC training simulator; designed and shipped a FastAPI service that parses trainee voice commands into structured JSON readbacks in real time.
- Benchmarked open-source LLMs against latency and instruction-following accuracy; selected Qwen3 via Groq API as the best fit for real-time voice interactions.
- Engineered a deterministic validation layer that enforces ICAO phraseology, callsign matching, and ATC range rules — overriding LLM output when safety constraints fail, ensuring training fidelity.
- Integrated a self-hosted Piper TTS server and connected the full voice-to-readback pipeline to a Unity frontend.

Cloud-Native URL Shortener on GCP, 2026
Cloud Applications Engineering, KFUPM

- Built a 15-Factor URL shortener on GCP with two FastAPI microservices (shortener + redirect) on Cloud Run, backed by Firestore and a Memorystore Redis cache, fronted by API Gateway.
- Provisioned the full GCP environment as modular Terraform (Cloud Run, Firestore, Memorystore, API Gateway, Secret Manager, Cloud Monitoring) with a make destroy && make restore flow that idempotently rebuilds the stack in minutes.
- Built a keyless GitHub Actions CI/CD pipeline using Workload Identity Federation (no service-account keys) that ships SHA-tagged images, deploys both Cloud Run services, and runs post-deploy smoke tests on every push to main.

CERTIFICATES

- GCP Associate Cloud Engineer ACE 11/2025 🔗
- AWS Certified Cloud Practitioner CLF-C02 08/2025 🔗
- Advanced Artificial Intelligence — KAUST Academy 02/2024

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, SQL

Cloud & Infrastructure: AWS (Bedrock Agents, Lambda, DynamoDB, API Gateway, SNS, CDK, IAM), GCP (Cloud Run, Cloud Functions, Pub/Sub, Firestore, Memorystore (Redis), Secret Manager, API Gateway, IAM), Terraform, Docker

Backend & Web: FastAPI, Next.js, React, REST APIs, WebSockets, OpenAPI/Swagger

GenAI: LLM integration, AWS Bedrock Agents, prompt engineering, agentic workflows, self-hosted/open-source LLMs

Tools & Practices: GitHub, GitHub Actions, Linux/Bash, Infrastructure as Code, CI/CD

Languages: English (Proficient), Arabic (Native)